Vibrato and Tremolo Preservation during Phase Vocoder Time-Stretching

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ABSTRACT

Phase vocoder time-stretching slows vibrato and tremolo proportionally to the amount of time-stretching, altering the character of the original performance. We present a method for preserving the original rates of sub-audio modulations during time-stretching by using a second-order spectral analysis of the phase vocoder representation. Each amplitude and instantaneous frequency channel of the phase vocoder analysis is analyzed with a long-term discrete Fourier transform (2LTDFT), in which sub-audio modulations appear as spectral peaks. These modulation peaks are removed prior to time-stretching and re-imposed at their original rate on the stretched result. The method is demonstrated on synthetic amplitude and frequency modulated signals as well as recorded saxophone, violin, and flute tones.

1. INTRODUCTION

Phase vocoder (PV) time-stretching affects the rate of vibrato and tremolo in musical sounds by slowing them proportionally to the amount of time-stretching. This behavior is for many applications an acceptable or preferred outcome. However, in many musical instrument performances, the particular vibrato and tremolo rates are important acoustic features that are critical to the musical perception of the performances. These sub-audio rate modulations, both in amplitude and frequency, are an important part of musical instrument identification, a performer's style, and musical expression. By preserving the original rates of vibrato and tremolo after a sound has been time-scaled, these important musical perception cues may be retained.

In this paper we review the current methods of vibrato and tremolo analysis in a spectral representation of sound, introduce the relevant background in second order Fourier transforms and long duration Fourier transforms, and develop a method of maintaining the original vibrato and tremolo rates of a recorded sound using second order analysis of the phase vocoder representation.

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2. EXISTING METHODS

Carl Seashore performed extensive study of musical vibrato in the 1930's, in which he defined and characterized vibrato and tremolo of western musical instruments and voice [1]. He characterizes vibrato as a 4 to 8 Hz pulsation of pitch, loudness, and timbre. Here we will separate vibrato as only a frequency modulation phenomenon and tremolo as an amplitude modulation phenomenon. When we wish to discuss any periodic or quasi-periodic low frequency modulation, whether amplitude, frequency or a combination of them, we will call them sub-audio modulations. Further discussion of the vibrato and tremolo characteristics of individual musical instruments can be found in Timmers and Desain [2].

Research into sub-audio modulation of musical sounds typically involves automated detection of the existence of vibrato and/or tremolo in a musical signal and estimation of the rate and extent of the modulation. Rossignol et al. suggest five methods of detecting and estimating the vibrato of a musical signal [3], including methods operating on the DFT spectrum as well as methods operating on the fundamental frequency track f_0, the first of which was originally developed by Herrera and Bonada [4].

Prior efforts to remove or modify vibrato from a recorded sound are generally based on sinusoidal analysis synthesis systems [5, 6]. Arfib and Delprat suggest a method of vibrato alteration based on DFT, cepstral, and Hilbert pitch tracking techniques [7, 8]. Their method decomposes the pitch curve calculated in the cepstral domain into a mean frequency and an oscillatory frequency. The oscillatory frequency is modified before recombination with the mean frequency and transformation out of the cepstral domain.

Maher and Beauchamp created a method of removing vibrato from vocal sound recordings based on the MQ sound representation [9]. Their system smoothes the frequency of sinusoidal analysis tracks to remove the vibrato, but does not allow for modification of vibrato rates or re-introduction of vibrato to time stretched sounds.

Most relevant to our work, Marchand and Raspaud [10] created a second order sinusoidal system in which each amplitude and frequency track of a sinusoidal analysis is in turn analyzed as a series of sinusoidal tracks. This method is used in a time-stretching algorithm which preserves the vibrato and tremolo of the recorded sound. As will be seen, the idea of a second order spectral representation will be used in our PV based system presented below.

3. RELATED DFT TECHNIQUES

3.1 Schumacher 2DFT Analysis

Robert Schumacher defined the 2DFT method of analyzing low frequency and sub-audio periodicities [11]. His method performs a single DFT along each of the time channels of the spectrogram of a sound. The amplitude spectrum of this second order DFT shows the spectrum of the low frequency and sub-audio periodicity of each channel of the input sound. Averaging the second order DFTs across all of the first order spectral channels produces a spectrum in which low frequency and sub-audio amplitude modulation of the original sound can be identified as spectral peaks. Schumacher's 2DFT analysis method will serve as part of the analysis, transformation, synthesis system developed here to maintain sub-audio modulation rates.

3.2 Long-Term Discrete Fourier Transform

The idea that a single Fourier Transform can be performed over long durations of time-varying sound in order to create novel transformations was formalized by Hammer and Sundt [12]. They call this transformation the long-term discrete Fourier transform or LTDFT. We will see in our sub-audio modulation method that applying the idea of the LTDFT in a higher order context helps us to model globally relevant vibrato and tremolo rates. In general, a technique that produces abstract sounds in the first order LTDFT domain produces a useful model of sub-audio modulation in a higher-order domain.

4. SECOND-ORDER PV SUB-AUDIO MODULATION SYSTEM

The phase vocoder typically employs a frame length of 512 or 1024 samples. Using a sampling rate of 44100 Hz and a frame length of 1024 samples, the lowest frequency that can be analyzed is $f = SR/N = 44100/1024 = 43$ Hz. Sub-audio frequencies are not captured explicitly as energy in low-frequency bins but as a pattern of changes across multiple spectral frames. A frame length sufficient to capture modulations as low as 4 Hz would be $N = 44100/4 = 11025$ samples. A phase vocoder analysis with such a long frame length will smear note onsets and transients under time-stretching. However, a phase vocoder with this frame length does model sub-audio modulations as spectral energy, and consequently PV time-stretching extends these frequencies at their original rates.

In order to maintain the frame rate of a traditional phase vocoder and analyze sufficiently long periods of sound to characterize sub-audio modulation, we analyze *each spectral channel* of a PV analysis with a single long DFT (LTDFT). In each of these "2LTDFT" spectra the sub-audio modulation can typically be seen as a prominent peak in the 4 to 8 Hz range. This method allows us to modify or remove the sub-audio modulation by manipulating this 2LTDFT peak. Our method operates by (i) removing the energy in 2LTDFT bins, (ii) PV time-stretching the unmodulated result, (iii) performing another 2LTDFT analysis of the lengthened sound, and (iv) imposing the shape of the original sub-audio energy on the 2LTDFT spectra.

4.1 Removal of Sub-Audio Modulation

The first step is a phase vocoder analysis. Windowed and overlapped FFT frames are calculated, the complex rectangular spectra are converted to amplitude and phase spectra, and the instantaneous frequency is calculated for each spectral bin with the phase difference method. We call the amplitude spectrum $|X_n(k)|$ and the instantaneous frequency spectrum $\Delta\theta_n(k)$ for time frame n and bin number k.

Instead of considering the spectral frame as the indivisible object of further analysis, we consider the *time-evolution* of each spectral channel as our indivisible objects for further analysis. In order to emphasize that the frame number n is the independent variable, the k amplitude channels are denominated $|X_k(n)|$, and the instantaneous frequency channels are notated $\Delta\theta_k(n)$.

Each second order spectrum of the amplitude channel is defined as,

$$\hat{X}_a(k,m) = \mathrm{DFT}\big(|X_k(n)|\big), \tag{1}$$

where $\hat{X}_a(k,m)$ is the complex valued spectrum of each amplitude channel k and frequency bins m. We call these spectra the "2LTDFT amplitude spectra" following Schumacher's nomenclature.

As the instantaneous frequency calculations are bounded by π and $-\pi$, the instantaneous frequencies must be unwrapped to create a continuous function before the second-order analysis. After frequency unwrapping, each second order spectrum of the frequency channel is defined as,

$$\hat{X}_f(k,m) = \mathrm{DFT}\big(\Delta\theta_k(n)\big), \tag{2}$$

where $\hat{X}_f(k,m)$ is similarly the complex valued spectrum of each frequency channel.

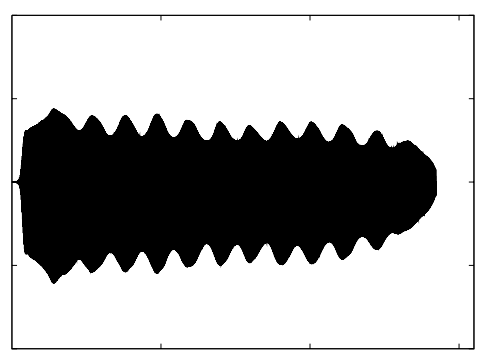

Figure 1. Saxophone sound showing amplitude modulation.

Figure 1 shows a time-domain signal of a saxophone note with prominent tremolo and vibrato. Visible in the 3 second sample is the slow amplitude variation at approximately 4 Hz.

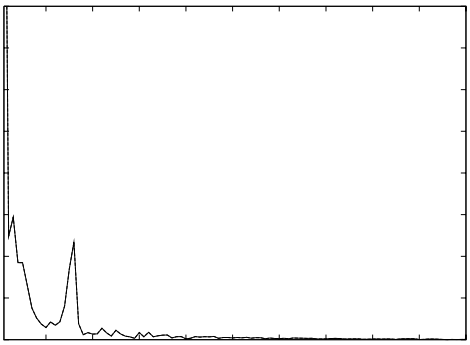

Figure 2. Amplitude of 2LTDFT amplitude spectrum of a single amplitude channel (channel 10).

Figure 2 shows the 2LTDFT amplitude spectrum of a single amplitude channel. A prominent spectral peak at approximately bin number 16 is visible. The center frequency for this bin is approximately 4 Hz. This spectral shape consisting of a large DC component and a prominent spectral peak around the vibrato and tremolo rate is characteristic of many musical instruments which feature this low frequency modulation.

We remove this low frequency modulation by "spectrally filtering" each of the 2LTDFT frequency and amplitude spectra. Our method consists of multiplying each 2LTDFT spectrum by an inverted Hamming window centered at the 2LTDFT bin with the maximum sub-audio frequency component. Figure 3 shows the 2LTDFT amplitude spectrum after the modulation peak has been removed.

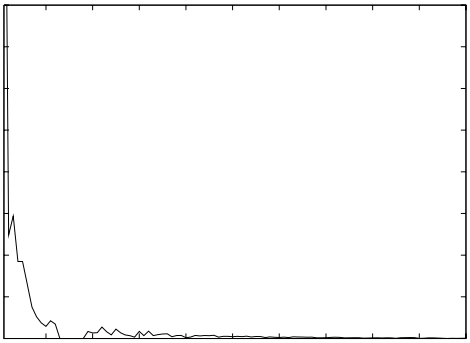

Figure 3. Amplitude of 2LTDFT amplitude spectrum after removal of modulation.

The original unmodified 2LTDFT bins are stored separately for use in re-introducing modulation after time-stretching. The set of 2LTDFT amplitude and frequency spectra are then each converted back to the first-order spectral representation by the inverse FFT.

4.2 De-modulated Lengthening

Next, a traditional phase vocoder time-stretch of this now "de-modulated" sound is performed. As the sound is already in a first-order phase vocoder representation, the initial steps of spectral analysis and instantaneous frequency calculation are already performed. All that is necessary is to calculate the new phase frames by phase advancement and calculate the amplitude frames by interpolation.

4.3 Re-introducing Modulation

The final step is to re-introduce sub-audio modulation at the original rate to our de-modulated and lengthened phase vocoder channels. Each amplitude and frequency channel is transformed into the second-order 2LTDFT format by taking the FFT of each channel across time. These new 2LTDFT spectra will be longer in proportion to the amount of time-stretching. Consequently, the original 2LTDFT sub-modulation bins cannot be directly combined into the new 2LTDFT spectra.

In order to add the original modulation rate to the 2LTDFT spectra, the original 2LTDFT bins representing the sub-audio modulation are interpolated to fit the corresponding spectral length in the new longer 2LTDFT spectra. This interpolated peak is substituted for the existing data in the new range of bins. Figure 4 shows these bins interpolated over the new longer 2LTDFT length.

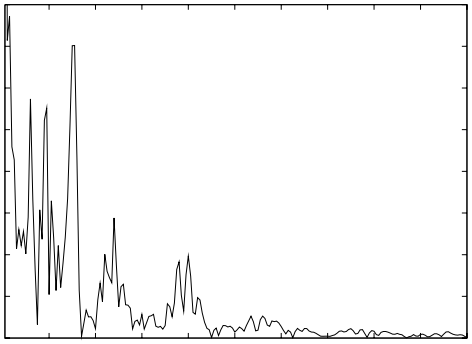

Figure 4. Amplitude of 2LTDFT amplitude spectrum with addition of interpolated modulation bins.

After substituting the new interpolated 2LTDFT values, the inverse FFT is performed on each spectral channel to return to the first-order phase vocoder representation. Phase accumulation followed by the standard phase vocoder inversion generates the new sound. This sound exhibits the long term development, pitch and timbre of the original sound slowed by the new factor but with the original tremolo and vibrato rates maintained.

Figure 5 shows the saxophone sound stretched to 6 seconds, where the original 4 Hz amplitude modulation rate is evident from visual inspection.

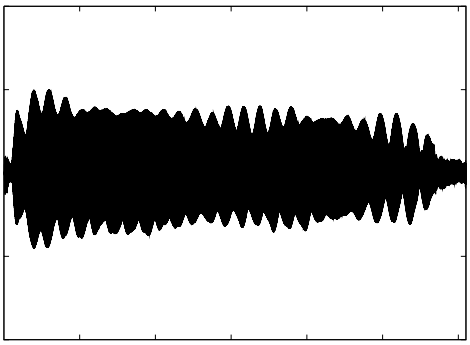

Figure 5. Time-stretched saxophone sound showing preserved amplitude and frequency modulation.

5. RESULTS

The algorithm was tested on several synthetic and musical signals with differing length, tremolo, and vibrato characteristics. All sound samples are monophonic and were sampled or generated at 44100 Hz. A Hamming window is used for the first-order FFT with a window size of 1024 and a first-order FFT overlap factor of 4. Sound examples are available at the repository accompanying this paper.

A 3 second synthetically generated amplitude modulated sine tone simulating a typical musical tremolo rate was generated with a 1000 Hz carrier frequency, a 5 Hz modulating frequency, and a 0.5 modulation depth. After removal of the modulation and stretching by a factor of 2, the original sub-audio modulation was re-imposed. In this example, the vibrato rate matches that of the original sound, but a time varying change in the vibrato can be heard. This variation consists of glitching artifacts at the beginning and end of the sound, and a gradual change in tremolo depth. Both problems are attributable to the phases of the 2LTDFT amplitude spectrum. When the phase spectrum of the 2LTDFT amplitude spectrum is altered by the addition of the interpolated bins, the evolution of the tremolo's depth is altered. The tremolo's frequency remains fixed because it is determined by the amplitude spectra of the 2LTDFT amplitude spectra. Attempts to alleviate this problem by randomizing the phases or leaving them unaltered resulted in sounds with less distinct tremolo characteristics.

The same procedure was applied to a frequency modulated sine tone simulating vibrato (1000 Hz carrier, 5 Hz modulating frequency, 0.5 modulation depth). The problems with the phase of the 2LTDFT spectrum also produce perceptually salient artifacts. The beginning and ending portions of the sound have no frequency modulation, but the central portion has the correct rate of modulation.

The same manipulations were applied to recorded saxophone, violin, and flute tones. For the saxophone and flute, both the vibrato and tremolo were removed and re-imposed on the lengthened version. Although the artifacts discussed above can be heard in the version with the modulation removed, the artifacts are much less apparent than in the sine wave examples when the vibrato and tremolo are re-imposed. This may be due to the relative complexity of the sound which masks the flaws in the process. For the violin, only the 2LTDFT frequency spectrum was manipulated. As with the saxophone example, the artifacts are less perceptible than in the synthetic examples.

6. CONCLUSIONS

We have shown that a method based on a higher order analysis of phase vocoder channel modulations can be used to alter the vibrato and tremolo characteristics of a sampled monophonic musical instrument sound. Our examples show that the method is able to retain the tremolo and vibrato rates of musical sounds with fairly stable vibrato and tremolo rates. The method achieves better results on sounds which have prominent tremolo and less well with sounds with prominent vibrato. This may be attributable to our ability to more easily perceive inappropriate pitch changes than amplitude changes. The method was unsuccessful at manipulating vibrato and tremolo on human singing tones due to the added complication of formant frequencies present in the human voice.

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